Muscle Pairs · Teacher and Cover Plan

Learning goal: I can find a pair of arm muscles and show which muscle pulls to bend or make the arm straight.

Scheme Outcome

BASE Science scheme of work, Unit 1 (Skeletons, Muscles and Nutrition), Topic: Identify that humans and some other animals have muscles for support and movement, and that muscles work in pairs by pulling on bones.

Identify that muscles work in pairs around a joint by pulling on bones while the partner muscle relaxes.

Success Criteria

I can point to the top and back muscles in an arm.

I can say that a muscle pulls on a bone and never pushes.

I can show which muscle pulls when an arm bends or goes straight.

Model Sequence (I do)

Look at Row 1. When the arm bends up, the top muscle gets short and pulls the bone.

At the same time, the back muscle stays soft and long so the arm can bend.

Look at Row 2. To go straight, the back muscle pulls and the top muscle rests.

A muscle is like a string or strap. You can pull with

a string, but you cannot push with it.

If both muscles pulled hard at once, the joint would lock still.

So one muscle must relax and go long while its partner pulls.

We Do & Hinge Check

When you bend your arm up towards your shoulder, what do your arm muscles do?

Answer: A. The top muscle pulls the lower arm bone up while the back muscle rests.

Safe Practical & Card Setup

Feel your own upper arm gently over your sleeve and match the arm pair cards.

Kit: Printed arm pair cards, Bend / Straight mats, and word strips.

Bend your own arm gently. Do not pull on anyone else's arm or lift heavy items.

You Do Routes (Supported, Independent, Aspire)

Supported: Match Pulls and Rests labels to the bent arm and straight arm pictures. (Help: Point to the fat, short muscle. That muscle pulls.)

Independent: Label the top and back muscles for a bent arm and a straight arm. Write one because sentence. (Help: When the arm bends, the __ muscle pulls and the __ muscle rests because __.)

Aspire: Explain what would happen if your arm had a

top muscle but no back muscle. (Help: The arm could bend up, but it could not go straight because muscles cannot __.)